FairCKD: Fairness Audit Report

Fairness-aware machine learning for automated chronic kidney disease diagnosis
fairckd.streamlit.app · Built by Jimil Jingar · PIET, Parul University

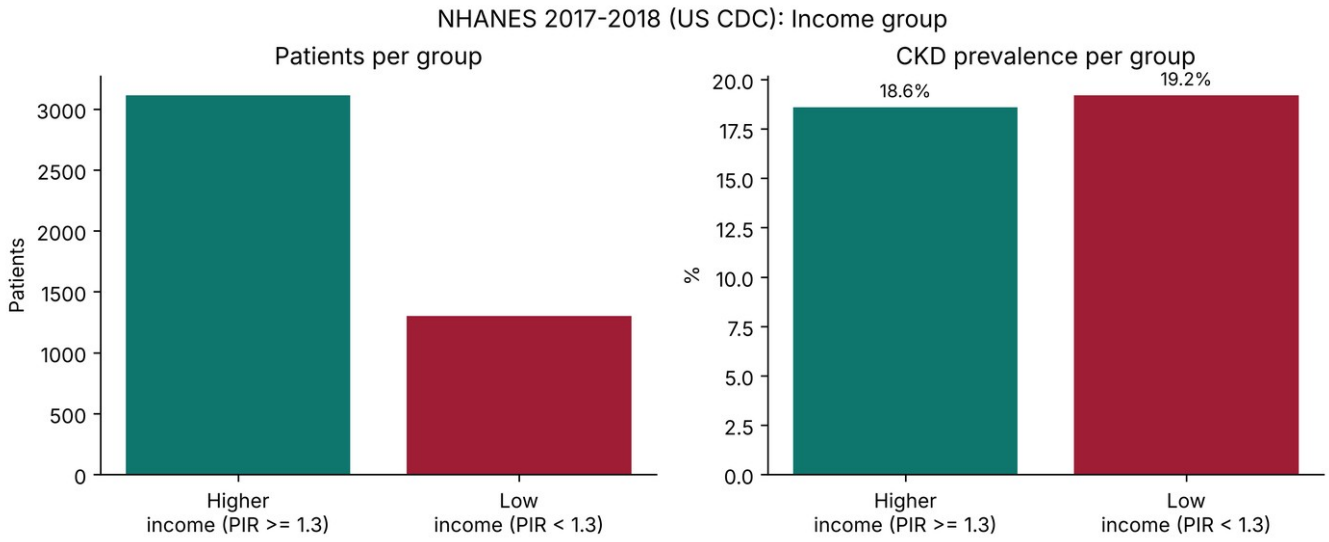
	Value
Dataset	NHANES 2017-2018 (US CDC)
Feature set	Routine blood tests (17 features, no kidney-function tests)
Sensitive attribute	Income group
Patients	4411
Features	17
Groups	Higher income (PIR >= 1.3), Low income (PIR < 1.3)
Decision threshold	training-set CKD prevalence
CV protocol	stratified 5-fold, 5 repeats (baseline), 5 repeats (mitigation)

Key findings

• Best baseline model: **XGBoost** (accuracy 0.774, balanced accuracy 0.738, AUC 0.803).
• Bias detection: in **7 of 7** models, CKD is detected least often in the **Low income (PIR < 1.3)** group (Equal Opportunity gap 0.036–0.095).
• Even a perfect classifier has DP diff = 0.006 and DI = 0.969 here, because CKD prevalence differs between groups; error-rate metrics (Equal Opportunity, Equalized Odds) are the clinically meaningful fairness criteria.
• Logistic Regression: best mitigation (balanced accuracy loss ≤ 3 points) = **Threshold Opt.**, Equalized Odds gap 0.066 → 0.011, balanced accuracy 0.736 → 0.725.
• XGBoost: best mitigation (balanced accuracy loss ≤ 3 points) = **Threshold Opt.**, Equalized Odds gap 0.089 → 0.017, balanced accuracy 0.738 → 0.711.
• **FairCKD-Select** (proposed, nested CV): Equalized Odds gap 0.073 → **0.007** compared with accuracy-driven model selection, balanced accuracy 0.737 → 0.729.
• Statistical significance: 0 of 13 mitigation results reduce the Equalized Odds gap significantly (paired bootstrap, p < 0.05).
• Proxy audit: the patient's group can be recovered from all model features with AUC **0.60** (0.5 = no group information).
• External validation (NHANES 2015–16): baseline balanced accuracy 0.744–0.747, Equalized Odds gap 0.026–0.030; best mitigated gap 0.009.
• Monitoring: 10 of 15 pipelines keep their Equalized Odds gap within 0.05 on the 2015–16 cohort; 3 feature(s) shift at least moderately within a group (PSI > 0.1).

1. Dataset and groups

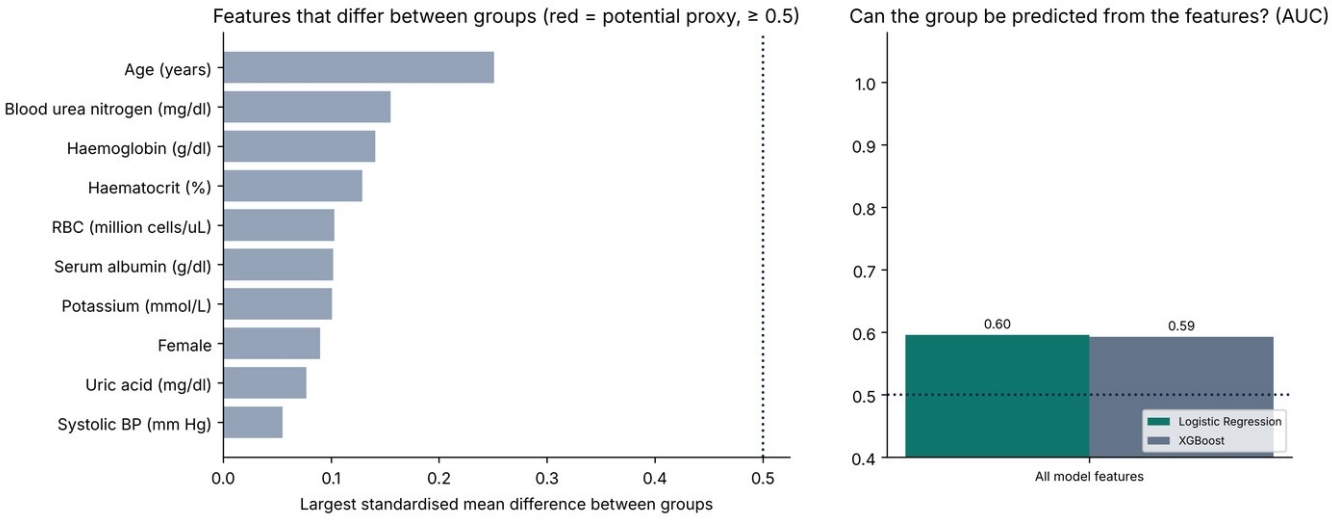
Group	Patients	Share (%)	CKD cases	CKD prevalence (%)	Missing values / patient
Higher income (PIR >= 1.3)	3112	70.6	578	18.6	0.120
Low income (PIR < 1.3)	1299	29.4	249	19.2	0.170
All	4411	100	827	18.7	0.140



1b. Proxy-feature audit

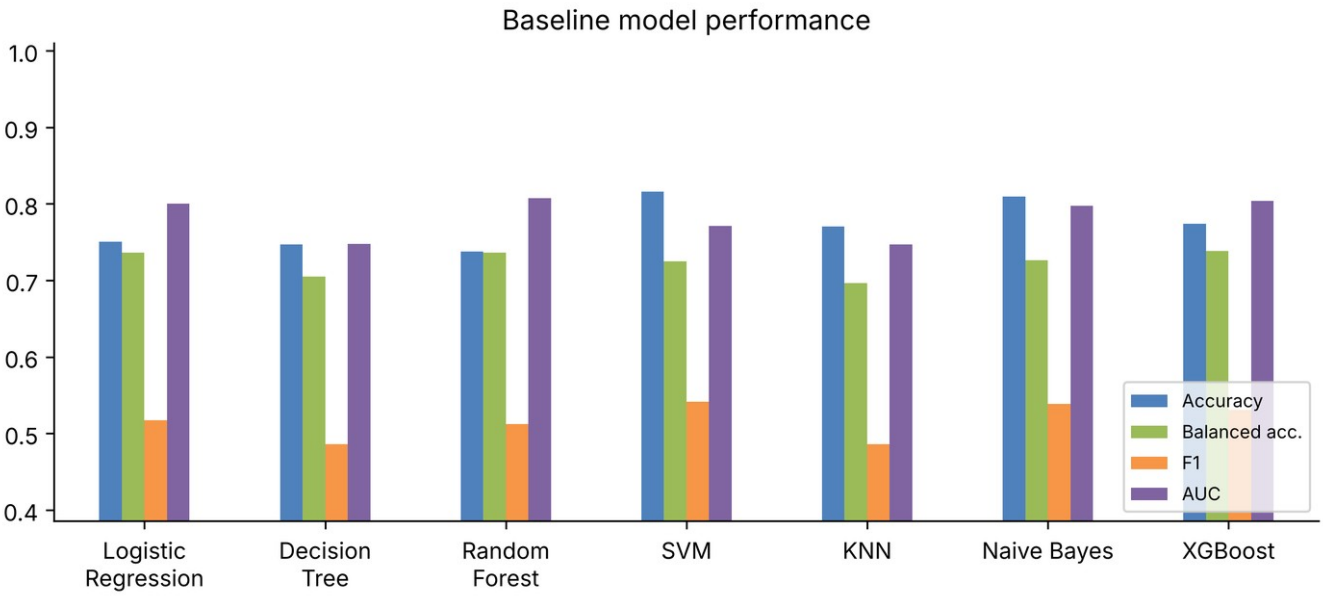
How much group information do the clinical features carry? A classifier is trained to recover the patient's group from the model inputs (5-fold cross-validated AUC; 0.5 = no information). A high AUC without the sensitive column means that simply dropping it ('unawareness') cannot make the model blind to the group.

Feature set	Features	AUC (Logistic Regression)	AUC (XGBoost)
All model features	17	0.596	0.593



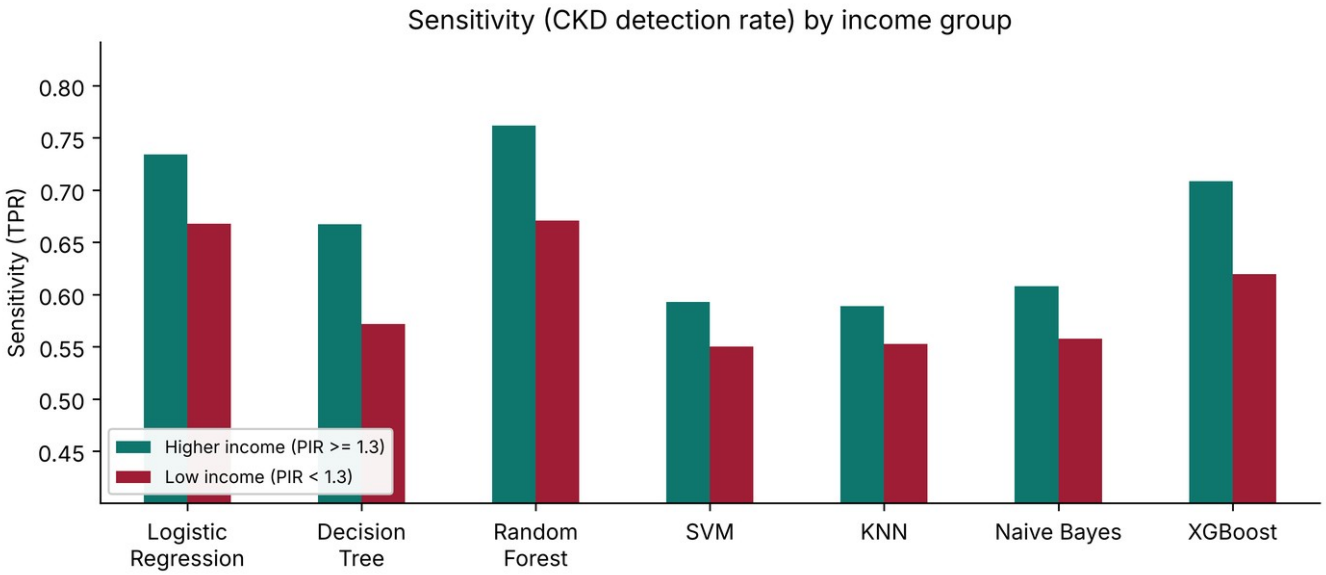
2. Baseline model performance

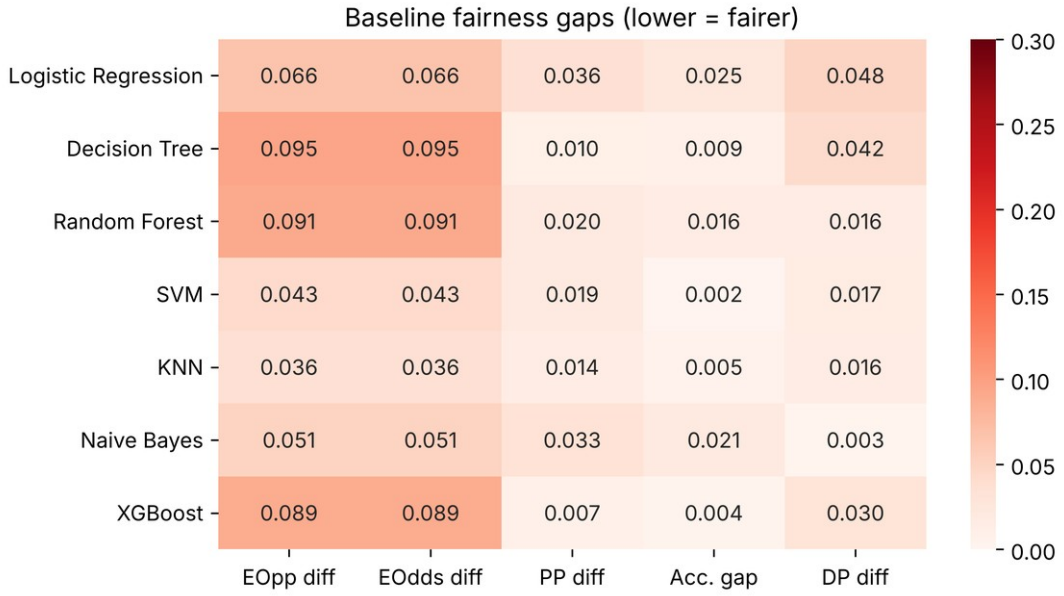
	Accuracy	Precision	Recall	F1	AUC	Balanced acc.
Logistic Regression	0.750 ± 0.001	0.406 ± 0.002	0.714 ± 0.002	0.517 ± 0.002	0.800 ± 0.001	0.736 ± 0.001
Decision Tree	0.747 ± 0.008	0.393 ± 0.009	0.638 ± 0.033	0.486 ± 0.012	0.748 ± 0.009	0.705 ± 0.011
Random Forest	0.738 ± 0.005	0.393 ± 0.005	0.734 ± 0.007	0.512 ± 0.005	0.808 ± 0.002	0.736 ± 0.004
SVM	0.816 ± 0.001	0.507 ± 0.003	0.580 ± 0.007	0.541 ± 0.003	0.771 ± 0.001	0.725 ± 0.003
KNN	0.770 ± 0.002	0.419 ± 0.005	0.578 ± 0.007	0.486 ± 0.005	0.747 ± 0.003	0.696 ± 0.004
Naive Bayes	0.810 ± 0.001	0.494 ± 0.003	0.593 ± 0.003	0.539 ± 0.002	0.798 ± 0.001	0.726 ± 0.001
XGBoost	0.774 ± 0.004	0.434 ± 0.007	0.681 ± 0.004	0.530 ± 0.005	0.803 ± 0.003	0.738 ± 0.004



3. Bias detection and fairness evaluation

	DP diff	DI ratio	EOpp diff	EOdds diff	PP diff	Acc. gap
Logistic Regression	0.048 ± 0.005	0.860 ± 0.014	0.066 ± 0.008	0.066 ± 0.008	0.036 ± 0.005	0.025 ± 0.004
Decision Tree	0.042 ± 0.018	0.871 ± 0.048	0.095 ± 0.025	0.095 ± 0.025	0.010 ± 0.007	0.009 ± 0.006
Random Forest	0.016 ± 0.012	0.956 ± 0.034	0.091 ± 0.014	0.091 ± 0.014	0.020 ± 0.014	0.016 ± 0.012
SVM	0.017 ± 0.006	0.923 ± 0.025	0.043 ± 0.013	0.043 ± 0.013	0.019 ± 0.006	0.002 ± 0.002
KNN	0.016 ± 0.005	0.938 ± 0.020	0.036 ± 0.015	0.036 ± 0.015	0.014 ± 0.008	0.005 ± 0.002
Naive Bayes	0.003 ± 0.002	0.987 ± 0.009	0.051 ± 0.008	0.051 ± 0.008	0.033 ± 0.009	0.021 ± 0.004
XGBoost	0.030 ± 0.008	0.902 ± 0.027	0.089 ± 0.017	0.089 ± 0.017	0.007 ± 0.007	0.004 ± 0.005
Perfect classifier (reference)	0.006	0.969	0.000	0.000	0.000	0.000


Sensitivity (CKD detection rate) by income group
Sensitivity (TPR)
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.45
Higher income (PIR >= 1.3)
Low income (PIR < 1.3)
Logistic Regression
Decision Tree
Random Forest
SVM
KNN
Naive Bayes
XGBoost


Baseline fairness gaps (lower = fairer)
Logistic Regression 0.066 0.066 0.036 0.025 0.048
Decision Tree 0.095 0.095 0.010 0.009 0.042
Random Forest 0.091 0.091 0.020 0.016 0.016
SVM 0.043 0.043 0.019 0.002 0.017
KNN 0.036 0.036 0.014 0.005 0.016
Naive Bayes 0.051 0.051 0.033 0.021 0.003
XGBoost 0.089 0.089 0.007 0.004 0.030
EOpp diff
EOdds diff
PP diff
Acc. gap
DP diff
0.30
0.25
0.20
0.15
0.10
0.05
0.00

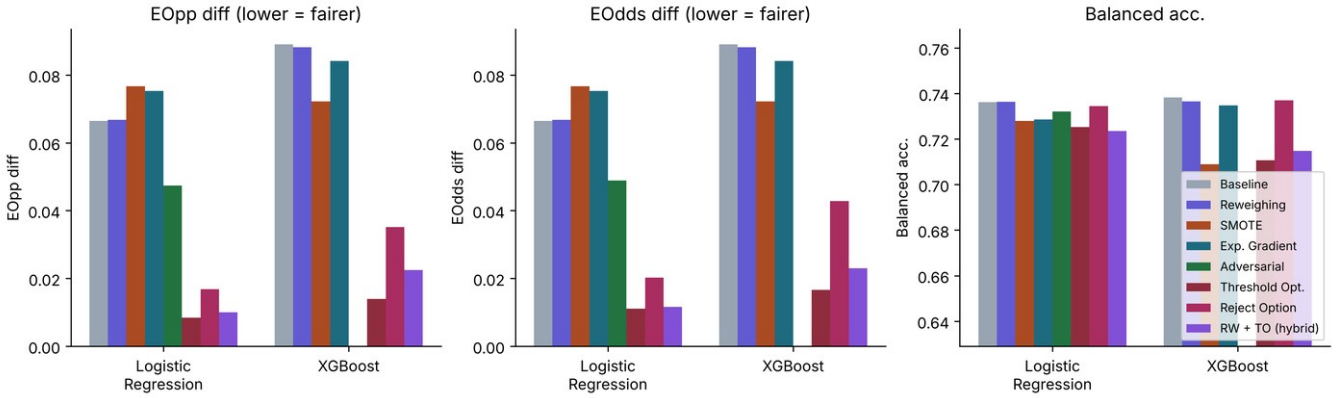
4. Bias mitigation

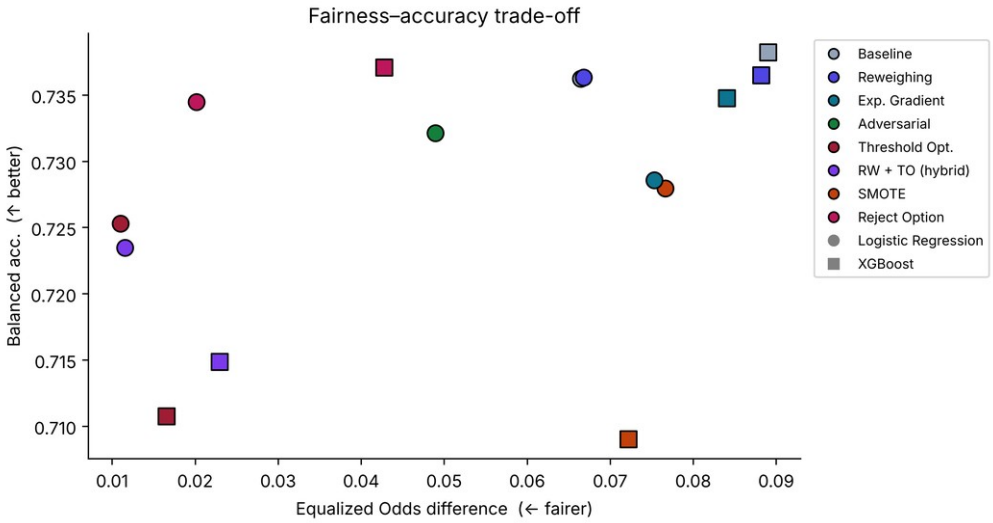
a) Performance (mean ± SD over CV repeats)

Model	Method	Accuracy	Balanced acc.	Recall	F1
Logistic Regression	Baseline	0.750 ± 0.001	0.736 ± 0.001	0.714 ± 0.002	0.517 ± 0.002
Logistic Regression	Reweighing	0.750 ± 0.001	0.736 ± 0.001	0.714 ± 0.002	0.517 ± 0.002
Logistic Regression	SMOTE	0.738 ± 0.001	0.728 ± 0.002	0.712 ± 0.004	0.505 ± 0.003
Logistic Regression	Exp. Gradient	0.750 ± 0.006	0.729 ± 0.009	0.694 ± 0.018	0.510 ± 0.011
Logistic Regression	Adversarial	0.749 ± 0.003	0.732 ± 0.004	0.705 ± 0.009	0.513 ± 0.005
Logistic Regression	Threshold Opt.	0.768 ± 0.009	0.725 ± 0.003	0.657 ± 0.016	0.515 ± 0.005
Logistic Regression	Reject Option	0.756 ± 0.002	0.734 ± 0.001	0.700 ± 0.003	0.518 ± 0.002
Logistic Regression	RW + TO (hybrid)	0.768 ± 0.008	0.723 ± 0.003	0.653 ± 0.015	0.513 ± 0.005
XGBoost	Baseline	0.774 ± 0.004	0.738 ± 0.004	0.681 ± 0.004	0.530 ± 0.005
XGBoost	Reweighing	0.775 ± 0.003	0.736 ± 0.003	0.674 ± 0.003	0.529 ± 0.004
XGBoost	SMOTE	0.829 ± 0.001	0.709 ± 0.002	0.516 ± 0.007	0.532 ± 0.002
XGBoost	Exp. Gradient	0.796 ± 0.005	0.735 ± 0.004	0.636 ± 0.006	0.539 ± 0.007
XGBoost	Threshold Opt.	0.772 ± 0.012	0.711 ± 0.002	0.613 ± 0.019	0.502 ± 0.006
XGBoost	Reject Option	0.779 ± 0.003	0.737 ± 0.003	0.670 ± 0.007	0.532 ± 0.004
XGBoost	RW + TO (hybrid)	0.786 ± 0.016	0.715 ± 0.003	0.600 ± 0.024	0.514 ± 0.010

b) Fairness gaps (lower = fairer; DI ratio: 1 = fair)

Model	Method	EOpp diff	EOdds diff	PP diff	DP diff	DI ratio	Acc. gap
Logistic Regression	Baseline	0.066 ± 0.008	0.066 ± 0.008	0.036 ± 0.005	0.048 ± 0.005	0.860 ± 0.014	0.025 ± 0.004
Logistic Regression	Reweighing	0.067 ± 0.008	0.067 ± 0.008	0.036 ± 0.006	0.048 ± 0.005	0.861 ± 0.014	0.025 ± 0.004
Logistic Regression	SMOTE	0.077 ± 0.007	0.077 ± 0.007	0.036 ± 0.009	0.056 ± 0.005	0.845 ± 0.014	0.029 ± 0.006
Logistic Regression	Exp. Gradient	0.075 ± 0.013	0.075 ± 0.013	0.018 ± 0.009	0.037 ± 0.011	0.889 ± 0.031	0.013 ± 0.006
Logistic Regression	Adversarial	0.047 ± 0.015	0.049 ± 0.013	0.024 ± 0.008	0.030 ± 0.004	0.910 ± 0.011	0.015 ± 0.005
Logistic Regression	Threshold Opt.	0.008 ± 0.007	0.011 ± 0.007	0.009 ± 0.004	0.008 ± 0.007	0.972 ± 0.024	0.005 ± 0.005
Logistic Regression	Reject Option	0.017 ± 0.008	0.020 ± 0.005	0.007 ± 0.007	0.009 ± 0.008	0.972 ± 0.023	0.012 ± 0.006
Logistic Regression	RW + TO (hybrid)	0.010 ± 0.007	0.012 ± 0.006	0.011 ± 0.006	0.010 ± 0.009	0.965 ± 0.030	0.006 ± 0.006
XGBoost	Baseline	0.089 ± 0.017	0.089 ± 0.017	0.007 ± 0.007	0.030 ± 0.008	0.902 ± 0.027	0.004 ± 0.005
XGBoost	Reweighing	0.088 ± 0.009	0.088 ± 0.009	0.007 ± 0.003	0.027 ± 0.006	0.910 ± 0.021	0.005 ± 0.004
XGBoost	SMOTE	0.072 ± 0.020	0.072 ± 0.020	0.044 ± 0.018	0.033 ± 0.005	0.825 ± 0.022	0.006 ± 0.005
XGBoost	Exp. Gradient	0.084 ± 0.014	0.084 ± 0.014	0.013 ± 0.012	0.032 ± 0.009	0.879 ± 0.036	0.005 ± 0.003
XGBoost	Threshold Opt.	0.014 ± 0.012	0.017 ± 0.009	0.012 ± 0.005	0.008 ± 0.005	0.971 ± 0.016	0.004 ± 0.004
XGBoost	Reject Option	0.035 ± 0.015	0.043 ± 0.012	0.042 ± 0.012	0.022 ± 0.011	0.927 ± 0.035	0.034 ± 0.009
XGBoost	RW + TO (hybrid)	0.023 ± 0.016	0.023 ± 0.016	0.008 ± 0.005	0.008 ± 0.004	0.970 ± 0.015	0.008 ± 0.005





c) Statistical significance (paired bootstrap vs. baseline, 1000 resamples; Δ = method – baseline)

Model	Method	Δ EOdds gap	95% CI (EOdds)	p (EOdds)	Δ Balanced acc.	95% CI (Bal. acc.)	p (Bal. acc.)
Logistic Regression	Reweighing	0.000	[0.000, 0.001]	1	-0.000	[-0.000, 0.000]	0.758
Logistic Regression	SMOTE	0.013	[-0.013, 0.037]	0.272	-0.006	[-0.013, 0.001]	0.084
Logistic Regression	Exp. Gradient	0.013	[-0.012, 0.038]	0.340	-0.001	[-0.007, 0.004]	0.660
Logistic Regression	Adversarial	-0.005	[-0.026, 0.019]	0.590	-0.005	[-0.011, 0.001]	0.132
Logistic Regression	Threshold Opt.	-0.040	[-0.082, 0.027]	0.228	-0.015	[-0.025, -0.005]	0.006
Logistic Regression	Reject Option	-0.036	[-0.059, 0.006]	0.090	-0.001	[-0.008, 0.004]	0.630
Logistic Regression	RW + TO (hybrid)	-0.039	[-0.088, 0.033]	0.300	-0.016	[-0.026, -0.007]	0.002
XGBoost	Reweighing	0.003	[-0.024, 0.031]	0.808	-0.002	[-0.010, 0.005]	0.574
XGBoost	SMOTE	0.012	[-0.044, 0.071]	0.710	-0.029	[-0.044, -0.016]	0.000
XGBoost	Exp. Gradient	0.003	[-0.031, 0.040]	0.868	-0.006	[-0.015, 0.002]	0.140
XGBoost	Threshold Opt.	-0.048	[-0.106, 0.041]	0.246	-0.024	[-0.036, -0.013]	0.000
XGBoost	Reject Option	-0.028	[-0.072, 0.035]	0.376	-0.001	[-0.007, 0.005]	0.838
XGBoost	RW + TO (hybrid)	-0.047	[-0.093, 0.031]	0.216	-0.025	[-0.036, -0.014]	0.000

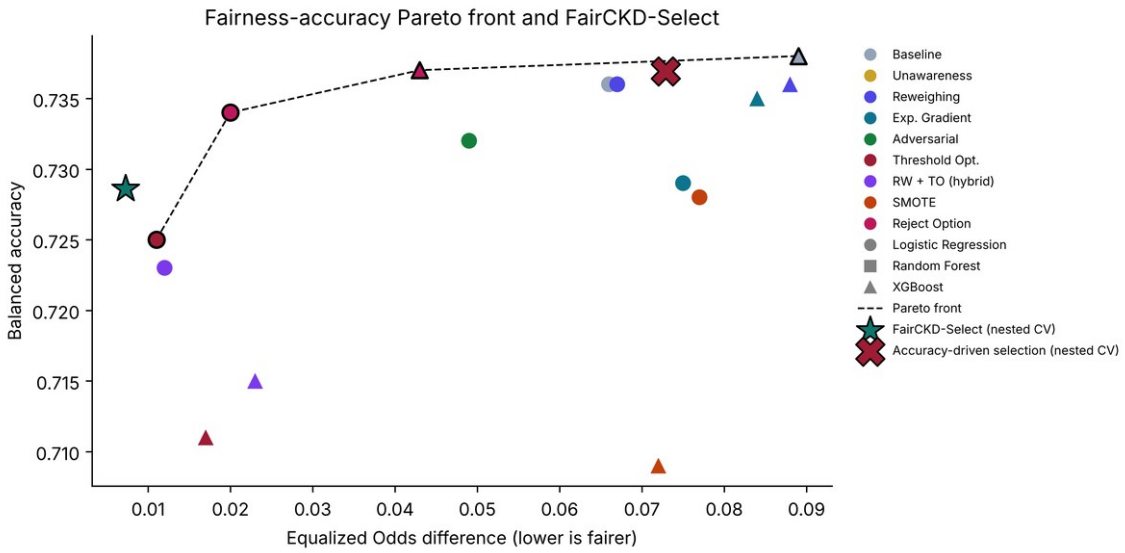
4b. FairCKD-Select (proposed method)

FairCKD-Select chooses, inside each training fold, the classifier and mitigation method with the smallest Equalized Odds gap among those whose balanced accuracy is within 3 points of the best unmitigated model (nested cross-validation). It is compared with conventional accuracy-driven selection.

Strategy	Accuracy	Balanced acc.	Recall	EOpp diff	EOdds diff	DP diff
FairCKD-Select	0.767 ± 0.004	0.729 ± 0.000	0.667 ± 0.005	0.007 ± 0.009	0.007 ± 0.008	0.001 ± 0.001
Accuracy-driven selection	0.757 ± 0.004	0.737 ± 0.001	0.706 ± 0.008	0.073 ± 0.011	0.073 ± 0.011	0.051 ± 0.006

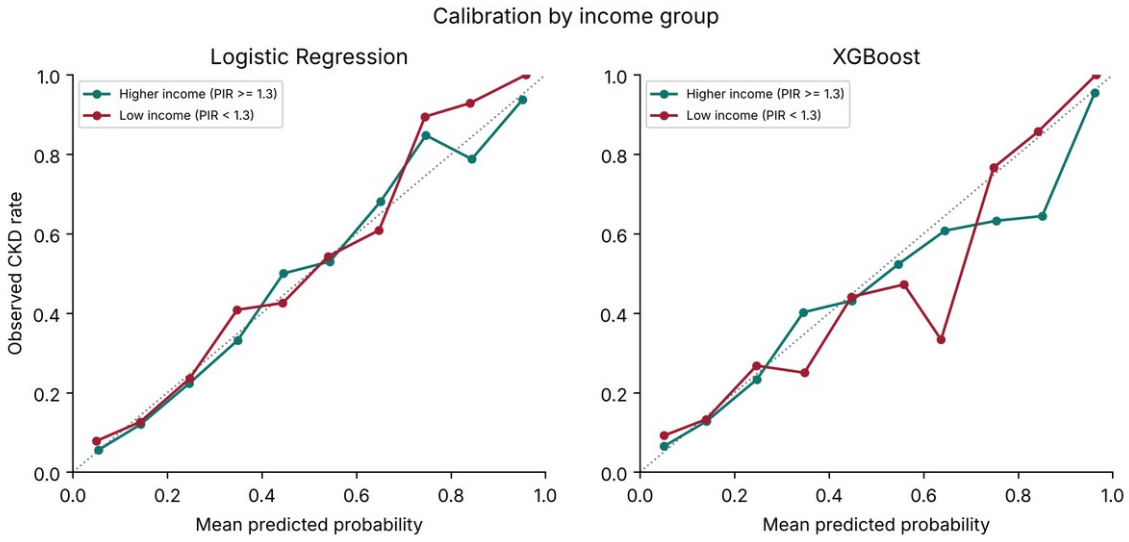
Pipelines chosen by FairCKD-Select across outer folds

Model	Method	Times selected
Logistic Regression	RW + TO (hybrid)	2
Logistic Regression	Reject Option	3
Logistic Regression	Threshold Opt.	2
XGBoost	RW + TO (hybrid)	2
XGBoost	Threshold Opt.	1



4c. Calibration by group

Group	Patients	Observed CKD (%)	Mean predicted (%)	Calibration-in-the-large (pp)	ECE	Brier	Model
Higher income (PIR >= 1.3)	3112	18.573	19.166	0.592	0.015	0.112	Logistic Regression
Low income (PIR < 1.3)	1299	19.169	17.585	-1.584	0.028	0.115	Logistic Regression
Higher income (PIR >= 1.3)	3112	18.573	18.584	0.011	0.022	0.112	XGBoost
Low income (PIR < 1.3)	1299	19.169	17.866	-1.303	0.039	0.118	XGBoost

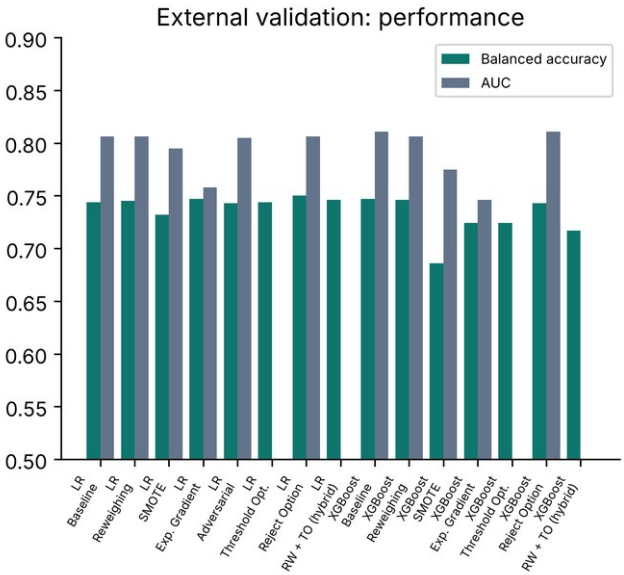


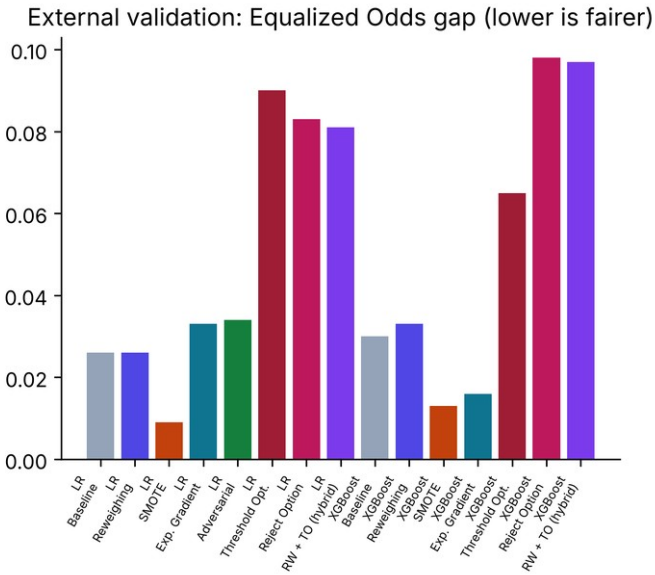
4d. Population-weighted estimates (NHANES survey weights)

Model	W Accuracy	W Balanced acc.	W EOpp diff	W EOdds diff	W DP diff	W TPR [Higher income (PIR >= 1.3)]	W TPR [Low income (PIR < 1.3)]
Logistic Regression	0.784	0.730	0.135	0.135	0.059	0.684	0.549
Decision Tree	0.787	0.701	0.125	0.125	0.046	0.607	0.482
Random Forest	0.783	0.742	0.138	0.138	0.025	0.715	0.577
SVM	0.848	0.713	0.085	0.085	0.025	0.543	0.458
KNN	0.804	0.696	0.084	0.084	0.019	0.562	0.478
Naive Bayes	0.849	0.719	0.089	0.089	0.005	0.557	0.468
XGBoost	0.808	0.728	0.137	0.137	0.046	0.644	0.507

4e. External validation (trained on NHANES 2017–18, tested on NHANES 2015–16)

Model	Method	Accuracy	Balanced acc.	Recall	AUC	EOpp diff	EOdds diff
Logistic Regression	Baseline	0.780	0.744	0.692	0.806	0.026	0.026
Logistic Regression	Reweighing	0.781	0.745	0.692	0.806	0.026	0.026
Logistic Regression	SMOTE	0.747	0.732	0.710	0.795	0.006	0.009
Logistic Regression	Exp. Gradient	0.772	0.747	0.710	0.758	0.033	0.033
Logistic Regression	Adversarial	0.778	0.743	0.693	0.805	0.034	0.034
Logistic Regression	Threshold Opt.	0.782	0.744	0.689		0.090	0.090
Logistic Regression	Reject Option	0.788	0.750	0.693	0.806	0.083	0.083
Logistic Regression	RW + TO (hybrid)	0.782	0.746	0.694		0.081	0.081
XGBoost	Baseline	0.795	0.747	0.676	0.811	0.030	0.030
XGBoost	Reweighing	0.797	0.746	0.671	0.806	0.033	0.033
XGBoost	SMOTE	0.836	0.686	0.465	0.775	0.013	0.013
XGBoost	Exp. Gradient	0.813	0.724	0.593	0.746	0.016	0.016
XGBoost	Threshold Opt.	0.827	0.724	0.573		0.065	0.065
XGBoost	Reject Option	0.799	0.743	0.661	0.811	0.098	0.098
XGBoost	RW + TO (hybrid)	0.816	0.717	0.571		0.097	0.097



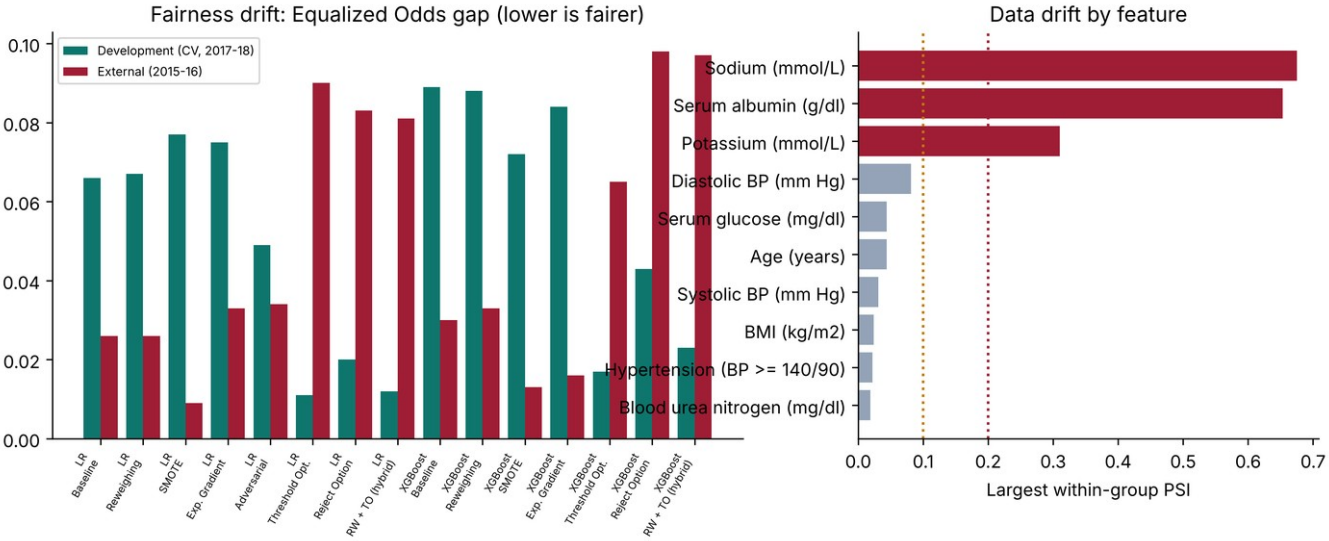


4f. Monitoring: data and fairness drift (development 2017–18 vs. 2015–16)

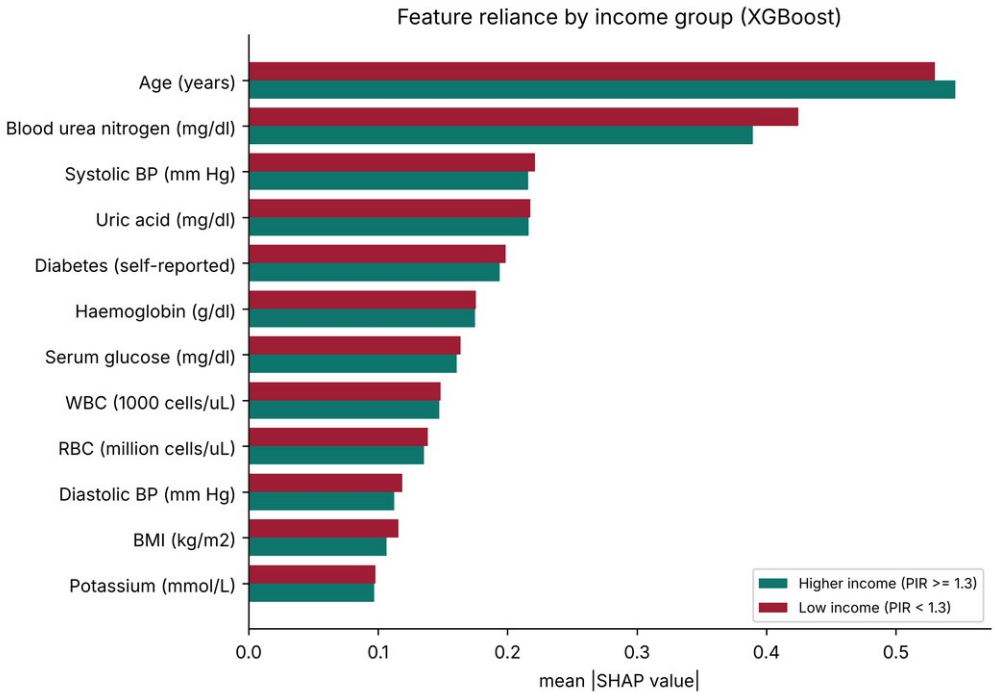
Population Stability Index (PSI) per feature and group (< 0.1 stable, 0.1–0.2 moderate, > 0.2 significant), CKD prevalence per group in both cohorts, and the change of each pipeline's Equalized Odds gap between the internal estimate and the external cohort.

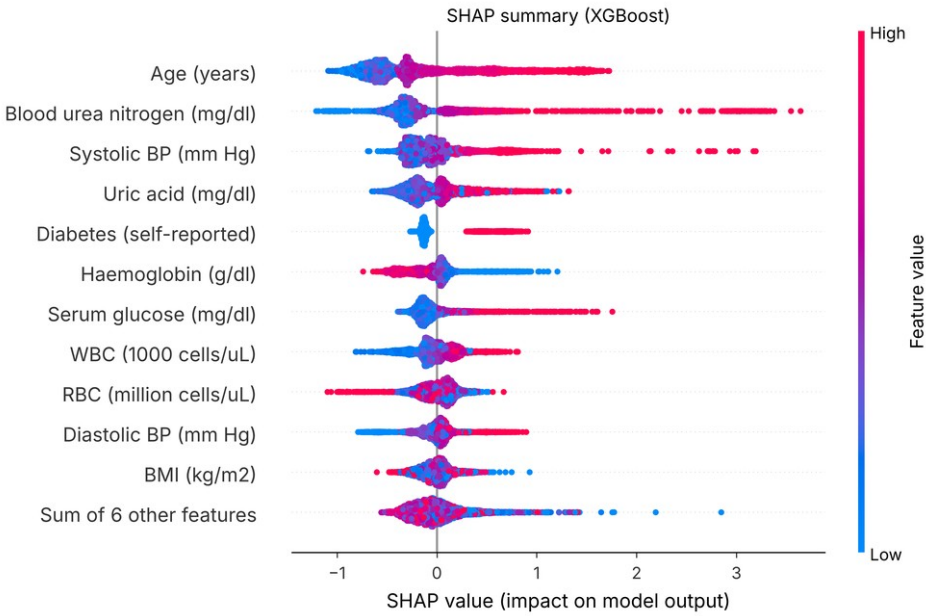
Group	Patients (development)	CKD % (development)	Patients (external)	CKD % (external)	Change (pp)
Higher income (PIR >= 1.3)	3112	18.57	3239	14.05	-4.53
Low income (PIR < 1.3)	1299	19.17	1536	19.99	0.820

Model	Method	Balanced acc. (internal)	EOdds diff (internal)	Balanced acc. (external)	EOdds diff (external)	Δ EOdds gap	Δ Balanced acc.
Logistic Regression	Baseline	0.736	0.066	0.744	0.026	-0.040	0.008
Logistic Regression	Reweighing	0.736	0.067	0.745	0.026	-0.041	0.009
Logistic Regression	SMOTE	0.728	0.077	0.732	0.009	-0.068	0.004
Logistic Regression	Exp. Gradient	0.729	0.075	0.747	0.033	-0.042	0.018
Logistic Regression	Adversarial	0.732	0.049	0.743	0.034	-0.015	0.011
Logistic Regression	Threshold Opt.	0.725	0.011	0.744	0.090	0.079	0.019
Logistic Regression	Reject Option	0.734	0.020	0.750	0.083	0.063	0.016
Logistic Regression	RW + TO (hybrid)	0.723	0.012	0.746	0.081	0.069	0.023
XGBoost	Baseline	0.738	0.089	0.747	0.030	-0.059	0.009
XGBoost	Reweighing	0.736	0.088	0.746	0.033	-0.055	0.010
XGBoost	SMOTE	0.709	0.072	0.686	0.013	-0.059	-0.023
XGBoost	Exp. Gradient	0.735	0.084	0.724	0.016	-0.068	-0.011
XGBoost	Threshold Opt.	0.711	0.017	0.724	0.065	0.048	0.013
XGBoost	Reject Option	0.737	0.043	0.743	0.098	0.055	0.006
XGBoost	RW + TO (hybrid)	0.715	0.023	0.717	0.097	0.074	0.002



6. Explainability (SHAP and LIME)

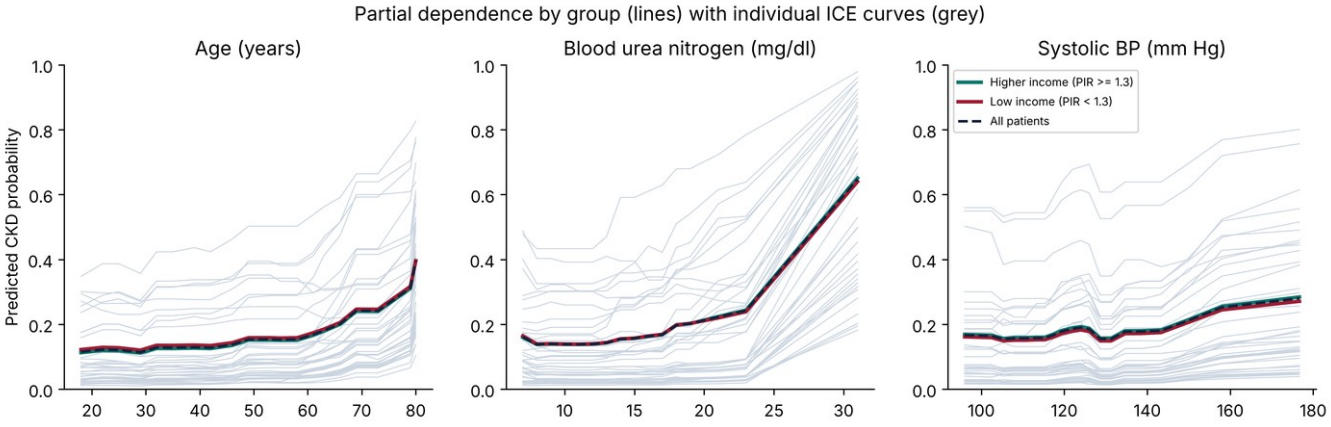




SHAP vs. LIME agreement: for a sample of patients per group, how many of the 5 most important SHAP features LIME also ranks in its top 5 (XGBoost). High overlap means the two explanation methods tell the same story.

Group	Patients	Mean top-5 overlap (of 5)	Mean LIME local R2
Higher income (PIR >= 1.3)	25	3.2	0.480
Low income (PIR < 1.3)	25	3.24	0.490
All	50	3.22	0.480

Partial dependence by group (top 3 SHAP features, XGBoost): the coloured lines show the average predicted risk in each group when the feature is set to the same value for everyone; grey lines are individual patients (ICE).



Protocol: repeated stratified 5-fold cross-validation (stratified on label × group); preprocessing and mitigation fitted on training folds only; mean ± SD over repeats. This system is a research prototype and not a medical device.